

CSC1109 – Data (Analytics) at Speed and Scale

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Things to Remember: Spark ML Part 1

Vectors such as `[1.0, 0.0, 3.0]` can be seen:

As a list of values (dense):

As non-zero indices and values (sparse):

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As a list of values (dense):

```
Vd = Vectors.dense(1.0, 0.0, 3.0)
```

As non-zero indices and values (sparse):

```
Vs = Vectors.sparse(3, [0, 2], [1.0, 3.0])
```

- Either can become a `labelledPoint` (by adding a label)

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- Supervised
 - Want to learn a function f so that $f(x) = y$
 - Have a lot of labelled $\langle x, y \rangle$ pairs
 - Classification and regression
 - Classification vs. regression?
 - Examples?
-

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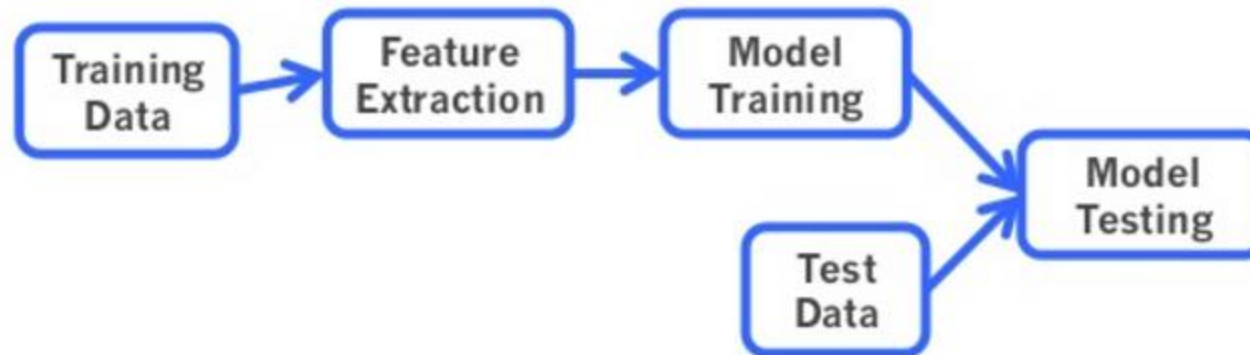
- Supervised
 - Want to learn a function f so that $f(x) = y$
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 - Classification vs. regression?
 - Examples?
- Vs. unsupervised
 - Now data is not labelled
 - Clustering algorithms (e.g....)

Things to Remember: Spark ML Part 2

- RDD vs Dataframes

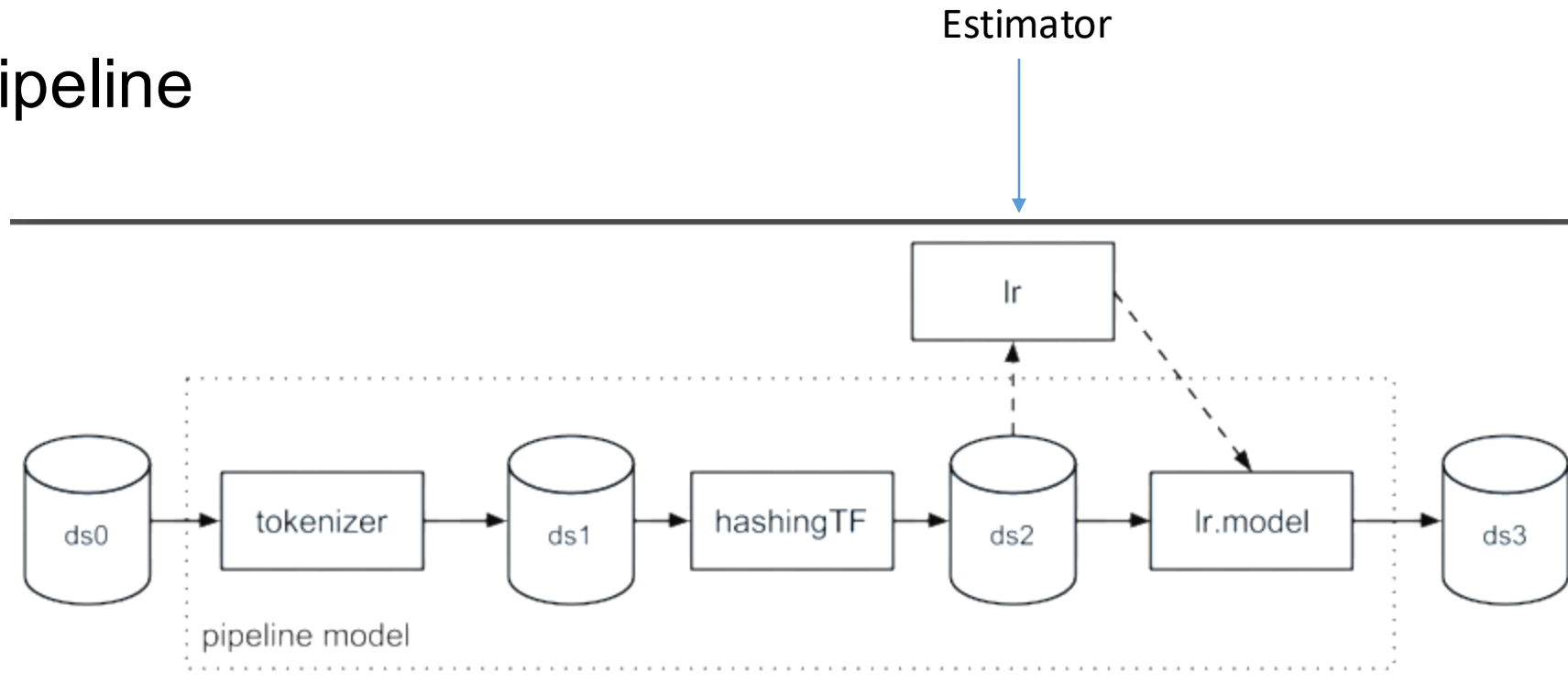
Things to Remember: Spark ML Part 2

- RDD vs Dataframes
- ML pipeline



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- RDD vs Dataframes
- ML pipeline (`spark.mllib` main guide, `spark.ml` package)
 - Not for RDDs (but you can still do ML with `spark.mllib` package)
 - Pipeline: prepare, extract ft, learn model, evaluate model, predict
 - Transformers – `transform()` vs. Estimator – `fit()`

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- ML pipeline (`spark.mllib` main guide, `spark.ml` package)
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 - Pipeline: prepare, extract ft, learn model, evaluate model, predict
 - Transformers – `transform()` vs. Estimator – `fit()`
- `spark.mllib(*)` vs `spark.ml(**)`
 - (*) <https://spark.apache.org/docs/latest/mllib-guide.html> vs <https://spark.apache.org/docs/latest/ml-guide.html>
 - (**) *Always check recent documentation vs. recordings*

